Grade 6 Accelerated　　Name____________________
Unit 10　　Date____________________
Lesson 1 Homework　　Period__________________

The Customer Service Manager at a call center collected data on the number of minutes a customer was on hold before speaking to a representative. Her data is shown.

8	12	11	7	8	10	9	35	9	10	14
13	4	15	12	10	21	6	5	15	13	11

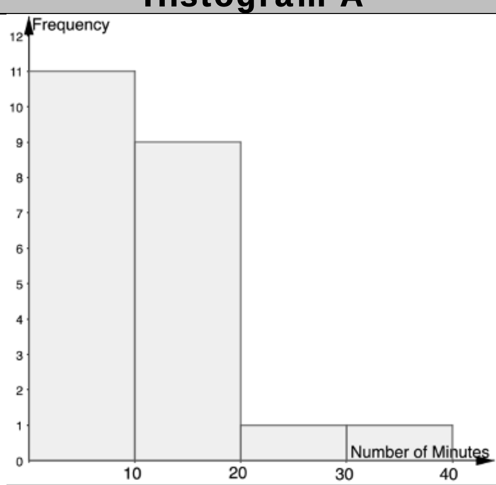
1. Create a frequency table of the data.

Number of Minutes	Frequency

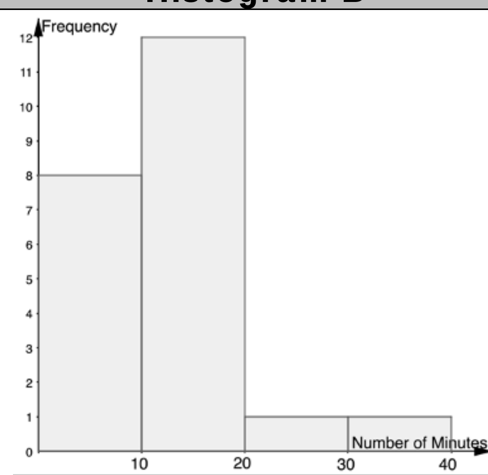
2. Determine the five-number summary of the data set.

Two histograms and two box plots are shown.

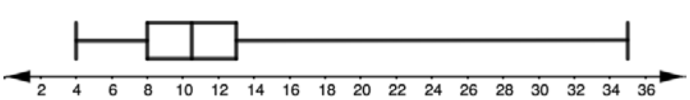
Histogram A



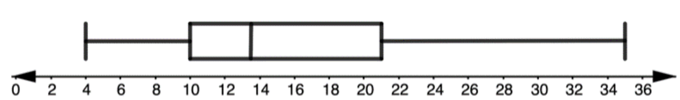
Histogram B



Box Plot A



Box Plot B

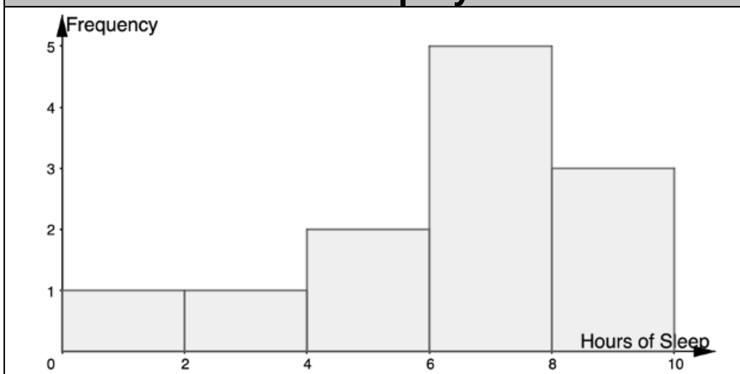
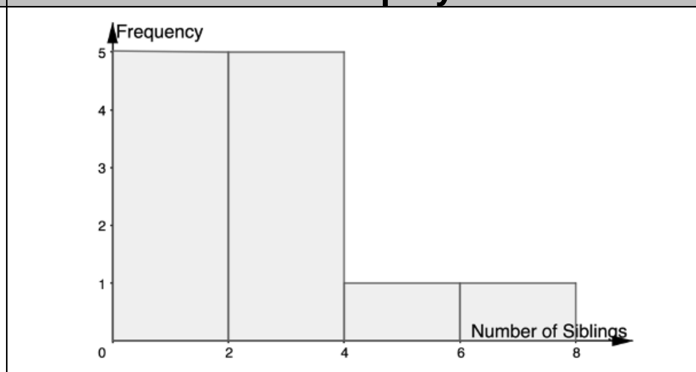
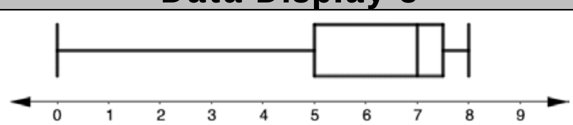
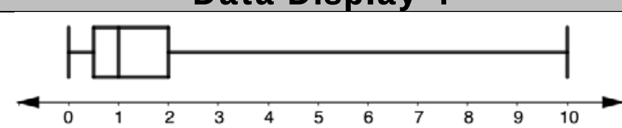


3. Explain which histogram better represents the data.

4. Explain which box plot better represents the data.

5. Which graph, box plot or histogram, is better for displaying this data? Explain your reasoning.

Four data displays and four data sets are shown.

<table>
<tr><th>Data Display 1</th><th>Data Display 2</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><th>Data Display 3</th><th>Data Display 4</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><th>Data Set A</th><th>Data Set B</th></tr>
<tr><td>0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 3, 10</td><td>0, 3, 4.5, 5, 6, 6.5, 7, 7, 7.5, 8, 9, 9.5</td></tr>
<tr><th>Data Set C</th><th>Data Set D</th></tr>
<tr><td>0, 0, 1, 1, 1, 2, 2, 2, 2, 3, 4, 6</td><td>0, 2, 4, 6, 6, 7, 7, 7, 7, 8, 8, 8</td></tr>
</table>

Match each Data Display to a Data Set.

6. Display 1: ___ 7. Display 2: ___

8. Display 3: ___ 9. Display 4: ___

10. Explain how you matched the data set that was represented by Data Display 3.

Grade 6 Accelerated Name___________________
Unit 10 Date___________________
Lesson 2 Homework Period__________________

Noelle kept track of the number of pages she read each day for the past 20 days. Her data are shown.

$$24, 35, 28, 27, 33, 36, 154, 28, 27, 25, 32, 25, 27, 31, 30, 28, 32, 25, 28, 30$$

1. One value in Noelle's data set is an outlier. Which value appears to be an outlier? Explain your reasoning.

2. Complete the statements.

   The mean of Noelle's full data set is _______.

   The mean of Noelle's data set with the outlier removed is _______.

3. Is there a significant difference between the mean of the full data set and that of the modified data set?

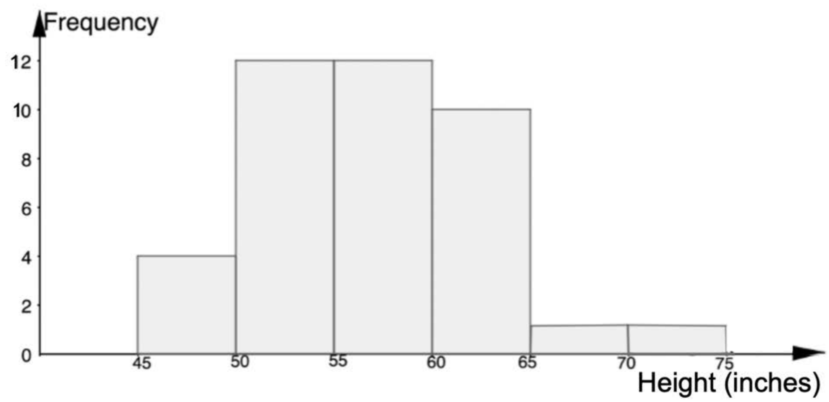
4. Complete the statements.

   The IQR of Noelle's full data set is _______.

   The IQR of Noelle's data set with the outlier removed is _______.

5. Is there a significant difference between the median of the full data set and that of the modified data set?

The histogram shows the heights, in inches, of students in a 6^{th} grade class.



6. Describe the shape of the histogram.

7. Explain which measure of center, mean or median, best represents this data.

8. Explain which measure of variability, range or IQR, best represents this data.

The length of the songs on an album, in minutes, is shown.

$$3.5, 3.5, 3.75, 4, 4, 4, 4.25, 4.5, 5, 5, 5.5, 5.75$$

9. Explain which measure of center, mean or median, best represents this data.

10. Explain which measure of variability, range or IQR, best represents this data.

Grade 6 Accelerated Name____________________
Unit 10 Date_____________________
Lesson 3 Homework Period____________________

1. Fill in the blanks with the terms provided.

mean	range	median	interquartile range (IQR)

The ___________ is the middle of an ordered list of values. It is a measure of center, or central tendency.

The ___________ is the arithmetic average of a set of numbers. It is a measure of center, or a central tendency.

The ___________ is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.

A measure of variation in a set of numerical data, the ____________ is the distance between the first and third quartiles of the data set.

The data for the number of emails a teacher gets on school days, Monday-Friday, was recorded for two weeks. Use the data shown for questions 2–8.

59	11	65	53	70
60	57	74	67	56

2. Determine the mean of the data set.

3. Determine the median of the data set.

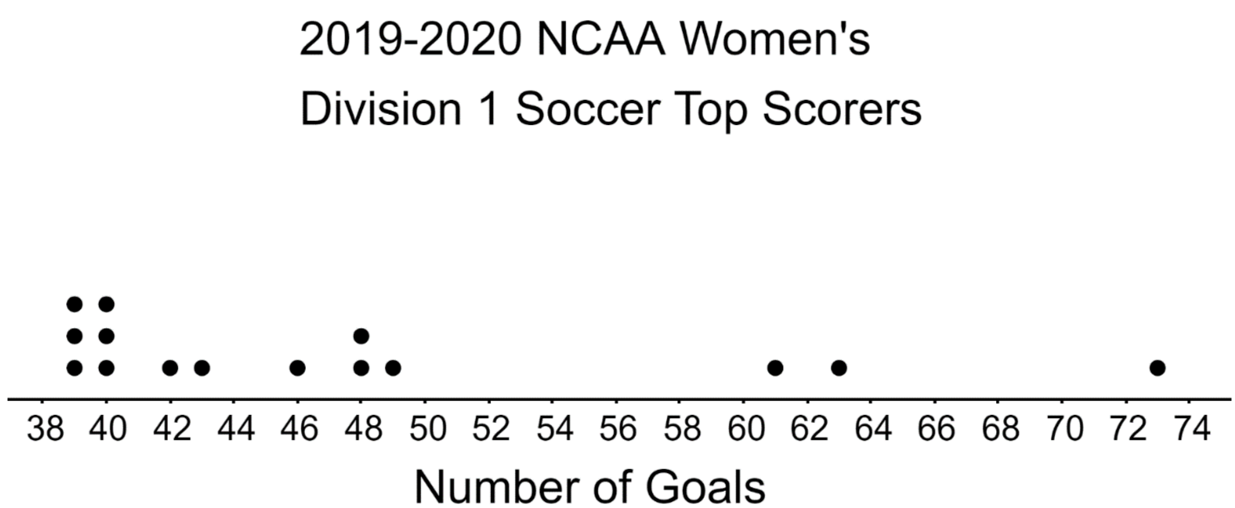
4. What is the range of the data set?

5. Which measures of center can be determined from a histogram?

6. The presence of an outlier does not have a significant effect on the IQR. True or False?

7. The presence of an outlier does have significant effect on the range. True or False

Below is a line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.



8. Which measure of the center, the mean or median, would be used summarize this data?

9. Explain whether the range or the interquartile range should be used to summarize the spread of the data

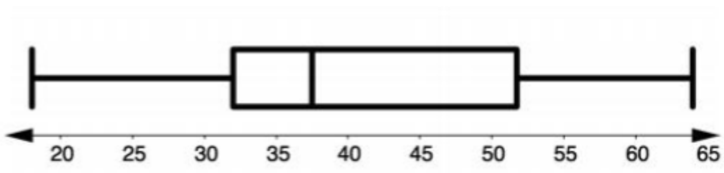
10. What is the range of the data?

Grade 6 Accelerated Name____________________
Unit 10 Date______________________
Lesson 4 Homework Period____________________

Mr. Ramirez is trying to determine the average score of his student's quiz grades on a recent quiz. He wrote down a summary of the student scores and displayed them in the stem-and-leaf plot and box plot.

Stem	Leaf
1	8, 9
2	5, 8
3	0, 0, 1, 2, 2, 6, 6, 6, 7, 7, 7, 8, 9
4	0, 1, 2, 8
5	1, 2, 5, 8, 8, 9, 9
6	0, 4

Key: 1|8 = 18



1. Which measure of center, the mean or median, best summarizes the quiz scores of the class?

2. Find the value of the measure you chose in question 1 if possible.

3. Describe the variation of quiz scores.

4. Does the range or the interquartile range better represent the variation in quiz scores?

Jackie is planning on buying a new car. She has a budget of $\$45{,}000$. The prices of new cars in her area are shown below, in thousands of dollars.

$$23, 52, 42, 38, 56, 70, 40$$

5. Which measure of center should Jackie use to summarize the prices of the new cars in her area, the mean or the median? Explain your reasoning.

6. Which measure of variation should Jackie use to summarize the prices of the new cars in her area, the IQR or the range? Explain your reasoning.

Jorge likes to listen to music on his phone. The data set below shows the times of the songs on his playlist.

$$3.25, 1.58, 2.22, 4.03, 3.57, 2.41, 7.46, 3.13, 4.01, 2.59$$

7. Are there any outliers in the data set? If so, which one?

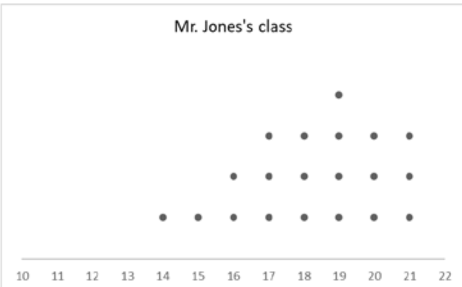
8. What measure of center best describes the typical song length on Jorge's playlist? Explain your reasoning.

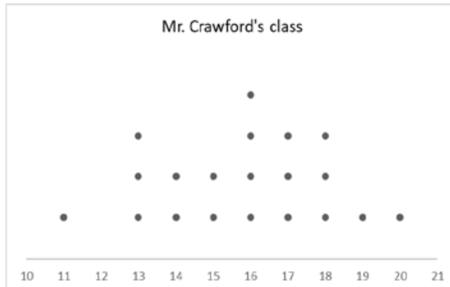
9. What measure of variation best describes the spread of the data distribution? Explain your reasoning.

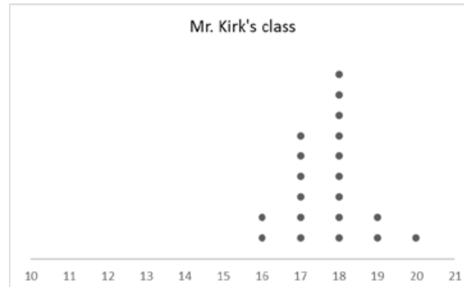
10. Jorge says the IQR is the most accurate depiction of the spread of his song lengths. Do you agree or disagree with Jorge? Explain your reasoning.

Grade 6 Accelerated		Name___________________
Unit 10				Date____________________
Lesson 5 Homework		Period___________________

The line plots below show the number of hours the students in three different classes spend watching videos on the internet each week.







Time (in hours)

1. Which line plot shown has a center of data that is greater than the center of data for Mr. Kirk's class?

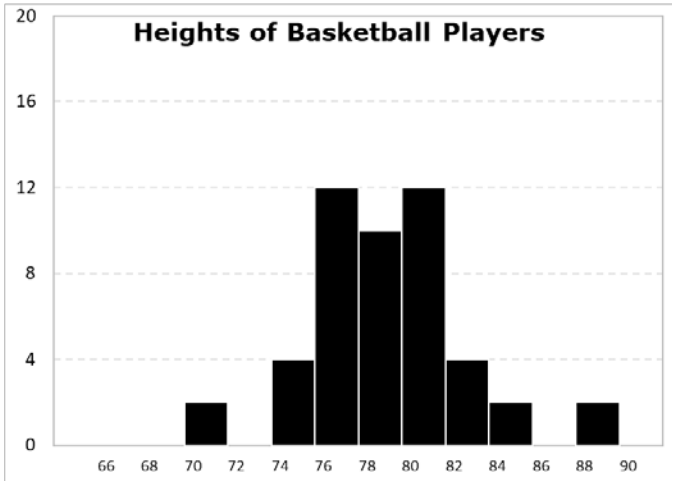
2. The value of the largest mean number of hours of all three classes is ________.

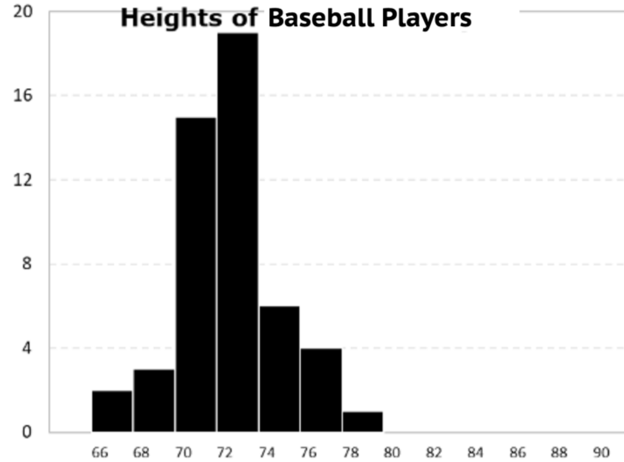
3. The value of the largest median number of hours of all three classes is ________.

4. Which of the line plots shown has the most variability in the times spent watching videos on the internet?

5. Based on the data from the line plots shown, which class spends the most time watching videos on the internet each week?

The two histograms show height distributions of 50 male professional basketball players and 50 male professional baseball players, in inches (in).





Height (in.)

6. The value of the larger mean height for the two groups is ________________.

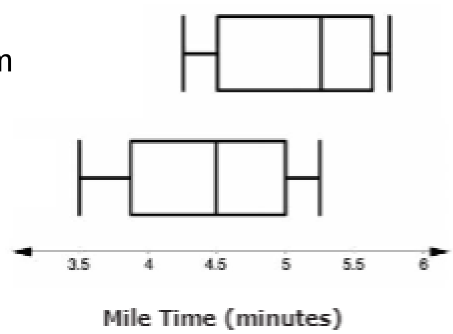
7. The value of the larger median height for the two groups is ________________.

8. In one to two sentences use the words shape, center, and variability to describe the differences between the two histograms.

The box plots shown below are the mile times for a JV and Varsity volleyball team.

JV Team

Varsity Team



9. Calculate and describe how the centers compare between the two teams.

10. Which team has the bigger variability? Explain your reasoning.

Grade 6 Accelerated Name___________________
Unit 10 Date____________________
Lesson 6 Homework Period___________________

A randomly selected group of male and female students were selected to have their heights measured at their middle school The heights, in inches, of the 15 randomly selected male and female students are shown below.

Male students: $56, 50, 49, 74, 46, 52, 61, 49, 53, 56, 60, 58, 47, 51, 59$

Female students: $40, 56, 50, 60, 66, 65, 55, 54, 52, 50, 49, 62, 61, 59, 64$

Complete table by using the heights of the male and female students shown above.

Measure	Male Students	Female Students
1. Mean		
2. Median		
3. Range		
4. IQR		
5. Potential Outliers		

6. Explain which measure you would use to describe the data sets above.

7. Which of the data sets have more variability?

A fast food restaurant was trying to determine which way customers could get their food the fastest, inside the lobby or in the drive thru. The restaurant paid 9 mystery shoppers to time how quickly they received their food in the lobby and in the drive thru. The times, in minutes are shown below.

Lobby	2	15	7	5	8	3	4	6	1
Drive Thru	5	3	6	5	4	6	2	7	9

8. What are the mean and median times for getting food in the lobby and in the drive thru?

9. Which method of getting food has the greatest IQR? Which method has the biggest range?

10. Based on the times shown above, use the most appropriate measures of center and variability to determine which method you would you chose to get your food from this restaurant.

Grade 6 Accelerated Name____________________
Unit 10 Date_____________________
Lesson 7 Homework Period____________________

A group of students challenged each other to see how long it took them to finish the same math problem. The students want to see who can solve the problem correctly the fastest. The times of the students are listed in seconds is represented in the stem and leaf plot.

Stem	Leaf
0	80
1	02 15 45 59 60
2	04 32 45
3	33

Key: $1|02 = 102$

1. Find the mean and median of the data set.

2. The range of the data set is _________________.

3. The IQR of the data set is __________________.

A different group of students in the same class timed themselves to see how long it took them to solve the same math problem and recorded their time in seconds.

$$\{134, 122, 111, 90, 101, 103, 143, 60, 45, 100\}$$

4. Find the mean and median of the data set.

5. The range of the data set is _________________.

6. The IQR of the data set is __________________.

For questions 7–10, determine whether each statement is true or false. Explain your reasoning.

7. The first group of students had a lower median than the second group of students.

8. The center of group 1 is similar to the center of group 2.

9. The second group of students has a larger range than the first group of students.

10. The first group of students is more consistent with their times than the second group.

Grade 6 Accelerated Name___________________
Unit 10 Date___________________
Lesson 8 Homework Period__________________

Scenario 1: Imagine a market researcher surveys 65 people on their coffee-drinking habits. He wants to know whether people in the local region are willing to switch their regular drink to something new.

1. In this scenario, what is the sample?

2. In this scenario, what is the population?

Scenario 2: Jasmine wants to collect data on the favorite dessert by young people in Jacksonville, FL. She decides to go outside a local store and ask every person who comes into the store during the morning and afternoon about their dessert preferences.

3. What is the population?

4. What is the sample?

5. Explain why a sample chosen in this way may not be representative of the population.

Scenario 3: John is going to sell lemonade outside his house and save money in order to buy a new laptop. He wants to predict his sales and estimate how many days he must sell lemonade. He is considering three different methods for surveying a sample. Describe the benefits and problems with each of the possible survey methods.

6. Ask everyone in his class if they like lemonade and how much will they drink in a day.

7. Ask all people younger than 15 around his house if they like lemonade and how much will they drink in a day.

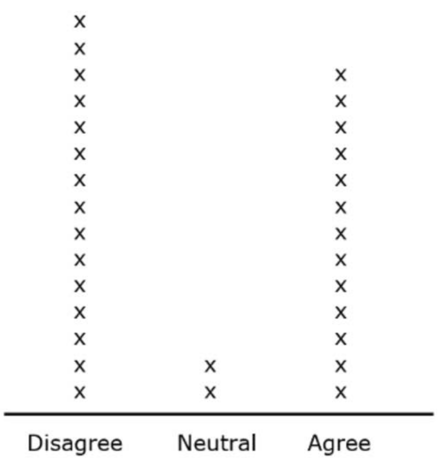
8. Ask all neighbors around his house if they like lemonade and how much will they drink in a day.

9. Explain which of the methods from questions 6, 7, and 8 would be the most likely to produce a sample that is representative of the population.

10. Determine a better way to select a sample for this study.

Grade 6 Accelerated		Name___________________
Unit 10				Date___________________
Lesson 9 Homework		Period__________________

A middle school in Warner Robins, GA is considering requiring students to wear uniforms for the upcoming school year. The decision was based on proposals by the school advisory committee and PTSA. Parents, teachers, and staff overwhelmingly agreed that wearing uniforms will have a positive effect on the school. The school's principal conducted a random survey of 30 students to gauge student opinions on the potential mandatory uniform policy. The results are shown on the line plot below. Use the data to answer questions 1-5.



1. What fraction of students surveyed were neutral on the potential mandatory school uniform policy?

2. A study conducted by a research group in Nevada indicated that 54% of middle school students agreed with a mandatory uniform policy. Explain how this compares to the students at the school in Warner Robins that agree with a mandatory uniform policy.

3. There are approximately $3{,}512$ middle students in Warner Robins, Georgia public schools. Predict the number of students in Warner Robins that would disagree with a mandatory uniform policy based on the data from the line plot.

4. There are approximately $3{,}512$ middle students in Warner Robins, Georgia public schools. Predict the number of students in Warner Robins that would be neutral about a mandatory uniform policy based on the data from the line plot.

5. Explain at least *two* factors that could contribute to the differences between your estimates and the actual number of students who could have voted about a mandatory uniform policy.

A guidance counselor at Top Performance High School in Washington, DC wants to analyze scores on the SAT to increase the average scores for seniors. The counselor gathered the SAT scores for all seniors at the school. The average SAT score in the US is 1051.

The counselor picked a random sample of 200 senior scores and recorded the information in the table below. Use this data to answer questions 6-10.

Students that scored under 1051	**Students that scored 1051 and higher.**
86	114

6. What is the proportion of students that scored 1051 or greater on the SAT at Top Performance High School?

7. Estimate the probability of a senior in the random sample scoring 1051 or greater on the SAT? Explain your method.

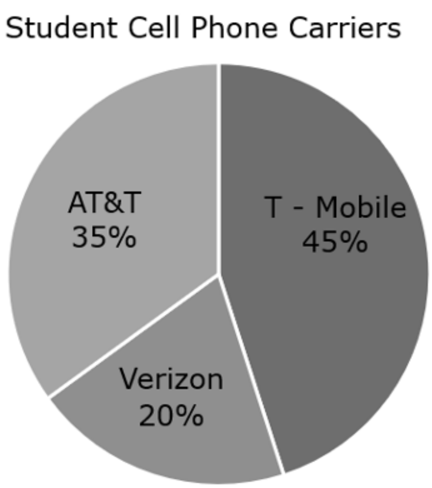
8. There are 638 seniors at Top Performance High School. Using equivalent fractions, how many seniors are expected to score 1051 or greater on the SAT (based on the sample data)?

9. If you use the percent method to answer question 8, what similarities and differences do you notice?

10. The guidance counselor estimates that 400 seniors will score 1051 or greater on the SAT sample data sample data. Explain whether this would be considered an unreasonable based on the sample data.

Grade 6 Accelerated   Name___________________
Unit 10   Date___________________
Lesson 10 Homework   Period__________________

550 Students in the 8^{th} grade were polled to see what cell phone carrier was the most common in a local middle school. The middle school has a total population of 2475 students. The circle graph below shows the percentage of students that have the three major cell phone carriers: AT&T, T- Mobile and Verizon.
Use the circle graph to answer questions 1-3.



1. To predict how many students with Verizon as a carrier, the student council president wants to use the scale factor method. What scale factor should she use to make this prediction?

2. Explain whether the equivalent factor or percent method be used to predict the number of students with AT&T as a carrier should the equivalent factor or percent method be used?

3. Predict the number of students that have T-Mobile as a cell phone carrier.

Tyrone wants to know about the color distribution of Skittles in his candy jar. Without looking he picked out one Skittle at a time, recorded it, and put them back into the jar. Tyrone picked 10 green, 6 purple, 12 red, 8 yellow, 4 orange after 40 picks. A total of 220 Skittles are in the jar. Use Tyrone's results to answer questions 4-5.

4. What fraction of Skittles are green, purple, red, yellow and orange?

5. Assuming his sample is a true representation of Skittle distribution, how can Tyrone predict the estimated number of Skittles in the jar that are red?

The cheerleading team at a local school cannot attend all of the games for all sports teams in the month of February due to conflicting schedules. The Athletic Director, Mrs. James, wants to prioritize the games the team will cheer at to boost morale of the teams. Mrs. James asks 150 students to pick the sport the cheerleading team should cheer. The results were recorded in the table below. The school has a total of 1150 students. Use the results of the sample to answer questions 6-10.

Sport	Number of Times Chosen	Ratio of Students that Picked Each Sport
Baseball	25	
Basketball	55	
Volleyball	40	
Soccer	30	

6. Find the ratio of students in the sample that picked each sport and complete the table.

7. Predict the number of students in the school that want the cheerleading team to perform at the baseball games (based on sample results) using the percent method.

8. Explain which is the best method to predict the number of students in the school that want the cheerleading team to perform at the soccer games (based on sample results).

9. Mrs. James uses the scale factor method to predict the number of students in the school that want the cheerleading team to cheer at basketball games. She uses a scale factor of 0.13. Did Mrs. James use the right scale factor? Explain.

10. The cheerleading team prefers to cheer at indoor sports. Predict the amount of students that prefer the cheerleading team to perform at the indoor sports.